Dirty business as coalition of the sidelined distort the clean energy debate

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It's time to call out the bluff and bluster of those who belligerently oppose private investment in renewable energy independent power projects (IPPs) and the transition away from coal and nuclear set out in South Africa's new electricity road map, the integrated resources plan (IRP). It is clear that some are frustrated and angry that they will no longer have access to special deals with Eskom or the coal and nuclear industries, others may feel marginalized and alienated from president Ramaphosa's new political dispensation – but their false narratives around IPPs and the IRP should not cloud our policy and investment choices.

The contestation around South Africa's energy future matters. The country can either embrace unprecedented global innovation in new low-cost energy technologies and competitive markets or it can succumb to a parochial and reactionary minority who are more interested in rent-seeking opportunities in old and expensive technologies that will impeded social and economic development. If the outcome is to be supportive of real economic transformation, hard facts and economic analysis must guide public debate and engagement.

Yet, it is a coalition of the recently sidelined and embittered that is currently dominating many public platforms, inciting resentment and opposition. An unlikely coalition, it includes elements from the Zuma-wing of the ANC, the EFF, actors with vested interests in nuclear energy and coal, such as the Nuclear Industry Association of South Africa and the Coal Transporters Forum, unions NUMSA and NUM, a small minority of energy professionals, including a number of ex-SOE board members and managers accused of corruption, the notorious Black Land First, and a collection of recently-formed advocacy groups such as Transform SA and the South Africa Energy Forum.

Bizarrely, this coalition of the disaffected has been joined on social media by individuals who deny anthropogenic climate change, as well as some members of the Free Market Foundation who apparently believe that government-backed coal and nuclear power projects, with a record of massive cost and time overruns, are preferable to South Africa's highly competitive IPP programme, which has attracted more than R200 billion in private investment in projects which have been built on-time and on-budget and, in the most recent bid round, without subsidies.

Their anti-IRP and IPP advocacy is founded on four fallacies: they argue that solar and wind energy are unreliable and expensive; that the IRP does not adequately take into account the impacts and costs of renewable IPPs on the rest of the electricity system; that these IPPs don't promote local industrialisation or create enough jobs; and that the procurement of IPPs has been corrupt and has benefitted the president's family and a new politically-connected elite. It is easy to demonstrate that each of these arguments is without substance.

While solar and wind energy were costly in the past, innovation and expanding global markets have led to dramatic cost reductions over the past five years and they are now the cheapest new sources of grid-connected power in South Africa. The most recent bid rounds in the renewable energy IPP procurement programme yielded prices of ZAR 62c/kWh, a third lower than Eskom's average cost of supply.

International renewable energy auctions are yielding even lower prices. For example, Egypt has solar contracts below US 3c/kWh and Mexico below US 2c/kWh. If South Africa ran a new competitive procurement now, solar and wind energy bids would be below ZAR 50c/kWh, lower than the operating costs of Eskom's most expensive coal power stations.

Ignoring these facts, some anti-IPP advocates point to countries with large shares of renewable energy where tariffs are high, for example, Germany and Denmark. But these countries were first movers, procuring renewable energy a decade ago when costs were much higher, and a large proportion of their retail tariffs are made up of taxes, making them unfit for comparison.

Others question the costs on the rest of the system for back up and ancillary power services, arguing that the IRP ignores these costs. In fact, the objective function of the IRP is the least-cost mix of power sources which delivers an acceptable reliability standard.

The PLEXOS computer model, which underpins the IRP, is the industry gold standard and is used by a large number of utilities and countries around the world. It shows that gas (or equivalent flexible resources such as hydro or batteries or demand-side management) complements the variability of solar and wind energy and, together, their weighted average cost is less than either coal or nuclear power stations - despite the fact that the IRP cost assumptions for solar and wind are conservative and those for nuclear optimistic.

The arguments of the anti-IPP coalition against the IRP simply do not stack-up and hence many in this group now abandon any pretence that nuclear or coal are least-cost. They argue instead that industrialisation and jobs should be the deciding factors in our new investment choices, ignoring the impact of electricity prices on affordability or economic growth.

While there is a need for more data and analysis, it is undeniable that renewable energy contributes to local manufacturing and jobs and has potential to do much more. The Department of Energy's data show that localisation in the renewable energy IPP programme is above 45%. Local equity ownership has exceeded 50% for many projects and more than 80% of debt financing has been from local banks. The benefits to South Africa are clear, though there may be room for policy adjustments that ensure that these benefits are distributed more equitably.

Disingenuously, some in the anti-IPP coalition point to the low number of operating jobs on a solar or wind farm compared to an Eskom power station, ignoring their vastly different energy outputs. On a like-for-like, jobs per kilowatt-hour produced comparison there is incontrovertible evidence that renewable energy generates more direct jobs than coal or nuclear energy. Another dishonest tactic is to compare the total of direct and indirect jobs of nuclear or coal with only the direct operating jobs of solar or wind farms. These strategies don't wash and are easily exposed.

Jobs in renewable energy involve basic construction, operation and maintenance skills. They are also hi-tech. A solar company in Cape Town not only does the engineering design for its international portfolio of projects (it's market share in solar PV is in the top ten globally) but its staff (nearly all South Africans) monitor and control, via computers and satellite communications, solar farms across the African continent, the Middle East, Asia and South America.

Finally, this disaffected coalition try to spread conspiracy theories that the IPP programme is corrupt and has benefitted members of President Ramaphosa's family and administration. Actually, the IPP procurement programme has been exemplary in its transparency, competitiveness and fairness. There have been more than 400 bids over five procurement rounds where around 92 projects have been awarded under conditions of fierce competition. Bid evaluations and awards have been undertaken under the strictest security, with independent and transparent auditing. The IPP procurement programme is, without doubt, the largest and most corruption-free public-private procurement undertaken by South Africa's government.

The propaganda techniques used by critics of the IPP programme mimic those of Trump. Inconvenient programme attributes are labelled fake news and 'alternative facts', repeated endlessly, seek to sway public opinion. Local use of Trumpian tactics is not coincidental. Members of this coalition have let slip their admiration of Trump's "strong-man" political style while ignoring his dismissal of Africa as "those shithole countries". They admire his promotion of coal and nuclear energy while ignoring data that shows the inexorable decline of both industries within the US. And their hypocrisy is further exposed as they profess adherence to radical economic transformation but fail to acknowledge that Trump's policies benefit the most wealthy one percent.

Facts and logic are probably not sufficient to win the argument for more private investment in renewable energy technologies. Nevertheless, Ramaphosa's administration will certainly need to expose the untruths and cynical agenda of those who seek to make the IRP and IPPs a political football.

It is insufficiently understood or acknowledged that IPPs currently contribute less than 5% of South Africa's electricity. Their costs are transferred directly to consumers through regulated tariffs. Eskom's finances or jobs are not yet directly impacted by IPPs. Neither are jobs in coal mines. But it is coming. Already work has started on a just transition where those most affected by job-losses in the future might be empowered by opportunities in new energy industries. Now is the time to chart out South Africa's energy path, not only in our energy-mix, but also in Eskom's structure and the growth of competition.

The economics of the energy transition to low-cost renewable energy, private investment and increased competition, are inexorable, just like they were in the ICT revolution in the 1990s. Some South African's may choose to fight this transition. That will set us back for a while. A small, backward-looking, previous political and professional elite may benefit briefly. But their children won't. President Ramaphosa and his Ministers now need to take the lead, and embrace the global tide of innovation in energy markets which can provide competitively priced, reliable and clean energy services for economic growth and prosperity for all our people.

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